

Replication Guide

1. Objective The purpose of this guide is to provide step-by-step instructions to completely reproduce the experimental results, statistical tests, and visualizations presented in the coursework report (specifically, Table 1 and Figure 1).

2. Environment & Data Setup

1. Ensure all dependencies listed in requirements.pdf are installed.
2. Ensure the datasets folder is placed in the same directory as the Python scripts. It must contain the four evaluated systems: xz, h2, dconvert, and x264.

3. Reproducing the Quantitative Results (Table 1 & Report Claims) To ensure reproducibility, random seeds have been hardcoded in the scripts (num_repeats = 30, train_frac = 0.7, random_seed = 1, and random_state=42 for the models).

1. Run python "Decision Tree.py".
2. Run python "Random Forest.py".
3. **To verify Table 1 (x264 System):** Scroll through the console output to find the dataset > System: x264, Dataset: blue_sky_1080p25_short.y4m.csv.
 - a) You will observe the baseline LR MAPE is ~0.7751.
 - b) In the Decision Tree output, the DT MAPE is ~0.1297 with a p-value of 1.8626e-09.
 - c) In the Random Forest output, the RF MAPE is ~0.1236 with a p-value of 1.8626e-09.
4. **To verify the artifcl.tar exception (Section 6 Reflection):** Find > System: xz, Dataset: artifcl.tar.csv. You will observe the console outputting: "Conclusion: Baseline model is better", verifying the specific case where Linear Regression outperformed the tree-based models.

4. Reproducing the Visualization (Figure 1)

1. After running the two scripts above, copy their respective terminal outputs.
2. Save the Decision Tree output as Decision tree results.txt in the same directory.
3. Save the Random Forest output as Random Forest results.txt in the same directory.
4. Run python Visualization.py.
5. The script will parse the .txt files using Regular Expressions and generate the system_comparison_3bars.png chart. The values in this generated image will perfectly match the values displayed in Figure 1 of the report.